

NICHOLAS SAROIU

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EDUCATION

- 2022-2026 **Honours Bachelor of Science**, Computer Science Major, Mathematics Minor
University of Toronto, Canada
Relevant Coursework: Neural Networks and Deep Learning; Introduction to Machine Learning; Computer Architecture; Operating Systems; Compilers and Interpreters; Parallel Programming; Introduction to Databases; Software Design.

WORK/VOLUNTEER EXPERIENCE

- SEPT 2025 **Student Researcher**
— PRESENT *Lee Language Lab* under supervision of Prof. Annie Lee, Toronto, Ontario
Conducting research on applications of Punctuation Restoration as an unsupervised learning objective for language models. Investigating language-agnostic performance gains, with a focus on improving model effectiveness for low-resource languages. Collaborating with a team of three students on experimental design and result analysis.
- MAY 2025 **Jr. Software Engineer**
— AUG 2025 *Dibbly Inc.*, Oakville, Ontario
Introduced automated validation frameworks (Jest, Pytest) to strengthen correctness and maintainability of the existing codebase. Contributed to development of full-stack AI applications within an agile team of ten developers using React.js, Next.js, and Python.
- JUNE 2024 **Frontend Developer**
— OCT 2024 Toronto, Ontario
Built an interactive real-time dashboard for residential battery systems displaying performance metrics and energy analytics. Developed responsive React.js components with Chart.js visualization to optimize energy consumption. Collaborated with independent dev team to deliver production-ready code.

PROJECTS

- OCT 2025 **Understanding Transformers via Causal Tracing**
Reproduced and analyzed interpretability experiments from the NeurIPS 2022 ROME paper "*Locating and Editing Factual Associations in GPT*". Applied causal tracing interventions on attention heads and intermediate activations to study how transformer models store and recall factual knowledge, using targeted ablation and restoration to identify influential internal components.
- NOV 2024 **Models to Predict Student Accuracy**
Built three machine learning models (K-Nearest Neighbors, Item-Response Theory, and Autoencoder Neural Network) to predict student performance. Implemented data preprocessing pipeline and ensemble methods to achieve 70% validation accuracy. Utilized PyTorch for neural networks and scikit-learn for traditional ML algorithms with cross-validation optimization.
- MAY 2024 **Patient Portal Backend**
Developed secure REST API for dental patient management system using Django REST framework and PostgreSQL. Implemented JWT authentication, data encryption for HIPAA compliance, and comprehensive CRUD operations. Created API documentation with Swagger.
- NOV 2023 **TTC Tracker**
<https://github.com/nsaroiu/TTC-Tracker>
Built real-time web app tracking Toronto transit vehicles with arrival predictions. Developed React.js frontend with interactive maps and Java Spring Boot backend integrating transit's GTFS data feed.

SKILLS

Python, Java, C, Javascript, Typescript, SQL, React.js, Next.js, PyTorch, scikit-learn, NumPy, Spring Boot, Django, Docker, Git.